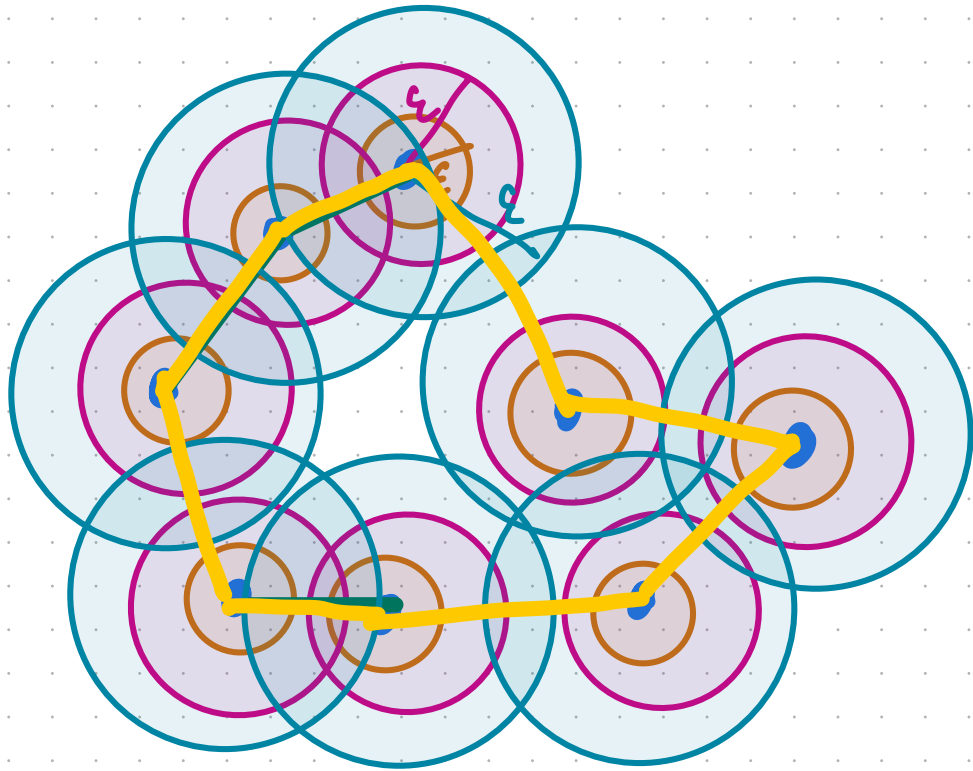


Week 5
2/18/26

TDA overview

Vietoris Rips complex (VR complex)

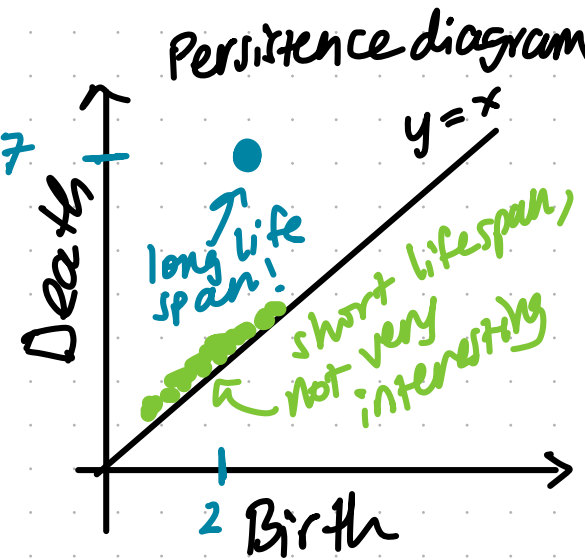
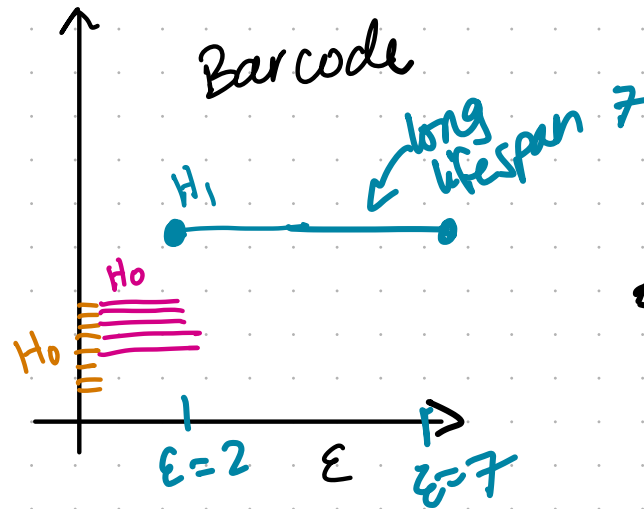


H_0 homology = connected component

H_1 homology = hole

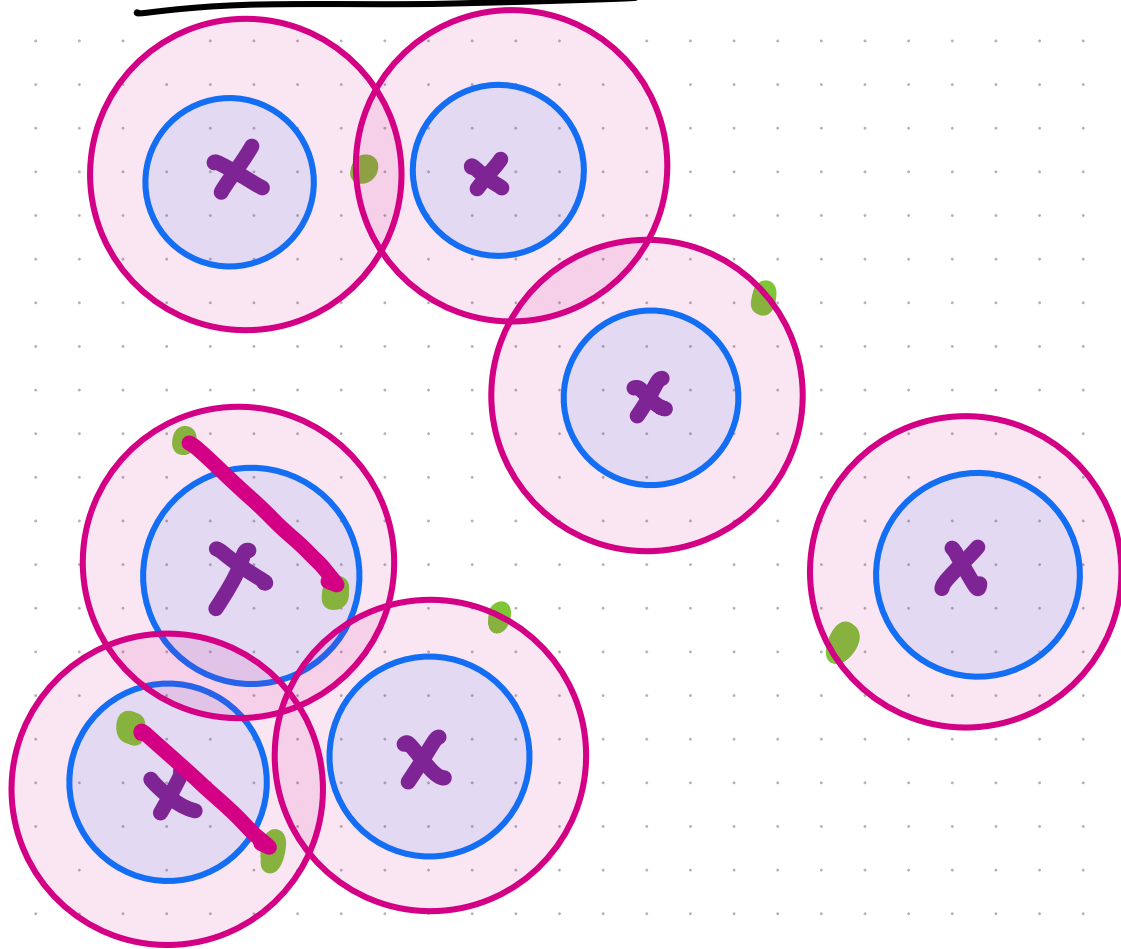
Pick ϵ radius $\rightarrow$ 8 separate components

$\uparrow \epsilon \rightarrow$ 5 separate components



Born at $\epsilon=2$
Died at $\epsilon=7$

Witness complex



● = landmarks

x = witnesses

Draw circles of radius ϵ around the witnesses (x's).

Connect two landmarks if they are within ϵ of the same witness.

landmark l of w in same w

$$\mathcal{W}(L, W)_\epsilon = \left\{ \sigma : \text{there exists } w \in W \text{ such that } d(l, w) \leq \epsilon \text{ for all } l \in \sigma \right\}$$

$\uparrow$ set of landmarks $\uparrow$ set of witnesses

L = centroids of census tracts

W = fire stations (for us)